Reducing Wait Times at Roots Natural Kitchen Using a Tandem Queueing Model

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Abstract*—Long lines at quick-service restaurants can reduce customer satisfaction and stress staff during peak demand. This paper models the in-person ordering process at Roots Natural Kitchen as a three-stage tandem queueing system: order taking, bowl assembly, and payment. Using real-time observations collected over three lunch periods, we estimate arrival and service parameters, validate the Markovian assumptions with Kolmogorov–Smirnov goodness-of-fit tests, and compare analytic steady-state predictions against a discrete-event simulation. The pooled model estimates an arrival rate of 0.662 customers per minute and identifies order taking as the bottleneck, with utilization 0.778. The analytic Jackson-network model predicts an average system time of 7.49 minutes, closely matching the simulation estimate of 7.45 minutes and remaining within roughly one minute of the observed average of 6.39 minutes. We then evaluate adding a second order-taker, which converts the first stage from M/M/1 to M/M/2. Erlang-C analysis predicts that this intervention reduces average system time to 3.60 minutes, a 52.0% reduction, while simulation gives a conservative estimate of 4.24 minutes. The results show that the first stage drives congestion and that targeted staffing at order taking offers the largest operational improvement.

***Keywords*—queueing theory, Jackson network, CTMC, discrete-event simulation, restaurant operations, Erlang-C**

I. Introduction

Roots Natural Kitchen is a quick-service restaurant near the University of Virginia that often experiences visible lunch-hour congestion. The customer experience is sequential: a customer joins the line, places an order, has the bowl assembled, pays, and leaves. Because each employee performs a specialized role, the system is not a set of interchangeable parallel servers. Instead, it is naturally modeled as a tandem service network.

This project was completed for the Stochastic Decision Models course at the University of Virginia, taught by Daniel Otero-Leon [1]. The course project asked teams to gather data, model a stochastic process, validate the model, and propose an improvement. Our goal is therefore both descriptive and prescriptive: identify where congestion forms in the Roots process and quantify whether a practical staffing change would reduce waiting time.

The main contributions of this paper are threefold. First, we formulate the Roots ordering process as a continuous-time Markov chain (CTMC) tandem network and explain why a single M/M/3 model would misrepresent the system. Second, we estimate the queueing parameters from timestamp data and validate the modeling assumptions using distributional tests and simulation. Third, we compare the base process against an improved design in which order taking is staffed by two parallel workers.

II. System and Data

A. Physical Process

The observed process has three service stages. Stage 1 is order taking, where the customer begins the interaction and communicates the desired bowl. Stage 2 is assembly, where toppings and ingredients are added. Stage 3 is cashier and payment. Customers must visit these stages in order; a worker at one stage cannot generally substitute for a worker at another stage.

For each observed arrival unit, the team recorded five timestamps: arrival to the line, start of Stage 1, start of Stage 2, start of Stage 3, and departure. The difference between departure and arrival gives observed total time in system. Interarrival times and service-stage durations were then computed from the cleaned timestamp sheet. The project repository, which includes the Excel workbook and analysis Jupyter notebook are included as project artifacts [2].

B. Data Collection Caveats

The data were collected in a crowded real environment with limited observers. When several customers appeared together, the team treated the batch as one arrival unit because individual customers in a group were difficult to separate reliably. The submitted data were also cleaned after collection to correct obvious timestamp-entry mistakes. These decisions make the data usable for a first queueing analysis, but they also mean the sample is not a perfect record of all individual-customer behavior. We return to these limitations in Section VII.

The final dataset contains 147 observed arrival units over three lunch observation windows: 50 on Day 1, 44 on Day 2, and 53 on Day 3. The pooled data therefore provide enough observations for a first-order stochastic model while still preserving day-to-day variation.

III. Queueing Model

A. Tandem CTMC Formulation

Let the CTMC state be

$$X(t) = (n_1(t), n_2(t), n_3(t)), \tag{1}$$

where $n_i(t)$ is the number of customers at stage i, including any customer in service. External arrivals enter Stage 1 at rate λ. Service completion at Stage 1 moves a customer from

Stage 1 to Stage 2 at rate μ_1 when $n_1 > 0$; completion at Stage 2 moves a customer to Stage 3 at rate μ_2 when $n_2 > 0$; completion at Stage 3 removes a customer from the system at rate μ_3 when $n_3 > 0$.

This is a tandem network of three M/M/1 queues. The M/M/1 assumption is that external interarrival times are exponentially distributed and that each stage's service time is exponentially distributed. In an open network of M/M/1 queues with Poisson external arrivals, the steady-state distribution has product form when $\rho_i = \lambda/\mu_i < 1$ for all stages. Each stage can therefore be analyzed with standard M/M/1 formulas:

$$\rho_i = \frac{\lambda}{\mu_i}, \qquad L_i = \frac{\rho_i}{1-\rho_i}, \qquad L_{q,i} = \frac{\rho_i^2}{1-\rho_i}, \tag{2}$$

$$W_i = \frac{1}{\mu_i - \lambda}, \qquad W_{q,i} = \frac{\rho_i}{\mu_i - \lambda}. \tag{3}$$

The total expected time in system is

$$W_{\text{sys}} = W_1 + W_2 + W_3. \tag{4}$$

B. Why Not M/M/3

An M/M/3 queue assumes three parallel, interchangeable servers drawing from one common queue. That is not the Roots process. At Roots, a customer cannot skip directly from the order-taking queue to payment, and the cashier is not an interchangeable substitute for the order-taker. Modeling the system as M/M/3 would therefore average away the serial structure and obscure the bottleneck.

IV. Parameter Estimation and Validation

A. Estimated Rates

For exponential interarrival and service times, the maximum-likelihood estimate of a rate is the reciprocal of the sample mean. Table I reports the estimated arrival rate, service rates, and utilizations by day and for the pooled sample.

TABLE I
Estimated Rates and Utilizations

Scenario	λ	μ_1	μ_2	μ_3	ρ_1	ρ_2	ρ_3
Day 1	0.585	0.833	2.142	1.130	0.702	0.273	0.517
Day 2	0.609	0.863	2.045	1.519	0.706	0.298	0.401
Day 3	0.825	0.860	2.279	1.360	0.958	0.362	0.606
Pooled	0.662	0.852	2.158	1.310	0.778	0.307	0.505

The pooled estimates identify Stage 1 as the bottleneck. Its utilization is 0.778, compared with 0.307 for assembly and 0.505 for payment. Day 3 is especially important because its estimated Stage 1 utilization is 0.958, which places the first stage close to the M/M/1 stability boundary.

B. Distributional Validation

To test whether the Markovian assumptions are reasonable, we fit exponential distributions to the interarrival samples and the three service-time samples. We then applied the Kolmogorov–Smirnov (KS) goodness-of-fit test. The results are shown in Table II and Fig. 1. All four samples fail to reject the exponential model at the 0.05 level. The coefficients of variation are also close to 1.0, which is the signature value for an exponential distribution.

TABLE II
Pooled Exponential-Fit Validation

Variable	n	Mean	Rate	CV	KS p
Interarrival	143	1.520	0.658	1.036	0.967
Stage 1 service	147	1.174	0.852	0.991	0.744
Stage 2 service	147	0.463	2.158	0.993	0.592
Stage 3 service	147	0.763	1.310	0.976	0.727

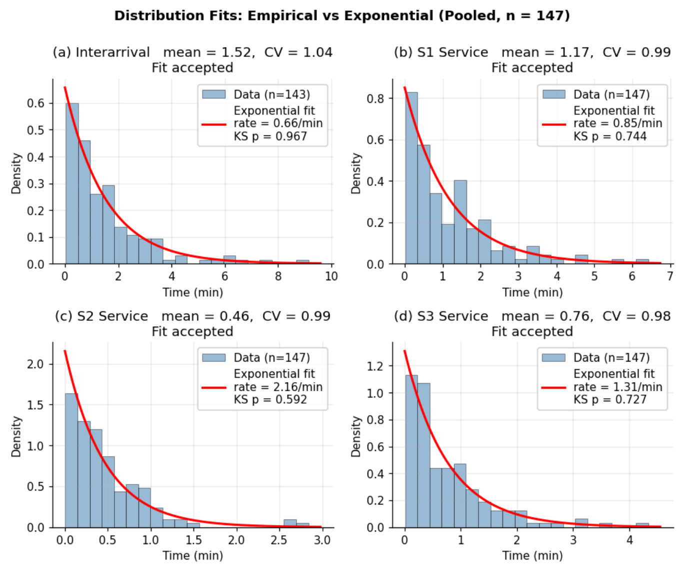

Fig. 1. Empirical interarrival and service-time distributions compared with fitted exponential distributions. All pooled KS tests fail to reject the exponential model at $\alpha = 0.05$.

C. Simulation Validation

Analytic formulas can hide implementation mistakes, so we validated the model using a discrete-event simulation (DES) of customer-by-customer movement through the three-stage tandem process. Each replication generates Poisson arrivals and exponential services using the estimated parameters, processes a large customer stream, discards an initial warmup period, and records mean time in system. Table III compares observed averages, DES output, and analytic predictions.

TABLE III
Observed, Simulation, and Analytic Mean System Time

Scenario	Observed W	DES W	Analytic W
Day 1	5.177 ± 0.422	6.473 ± 0.147	6.502
Day 2	4.802 ± 0.447	5.700 ± 0.152	5.731
Day 3	8.843 ± 0.621	29.931 ± 12.038	30.527
Pooled	6.387 ± 0.333	7.445 ± 0.234	7.490

The pooled analytic and DES estimates differ by less than 1%, which validates both the formula implementation and the simulation logic. Observed averages are lower than steady-state predictions because the collection windows are finite lunch intervals rather than long-run steady-state experiments. Day 3 also shows how sensitive M/M/1 waiting time becomes near saturation: as λ approaches μ_1, the term $1/(\mu_1 - \lambda)$ grows rapidly.

V. Base Model Results

The base model confirms that the first stage dominates system performance. For the pooled model, Stage 1 contributes 5.278 minutes of the 7.490-minute analytic system time. Stage 2 contributes only 0.669 minutes, and Stage 3 contributes 1.543 minutes. In other words, improving assembly would have little effect unless order taking is also improved.

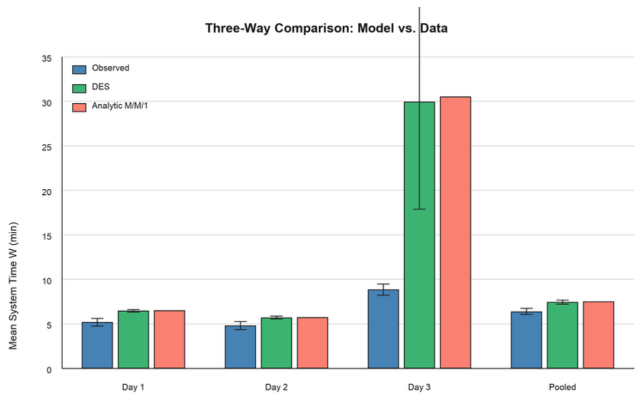

Fig. 2. Comparison of observed mean system time, DES output, and analytic M/M/1 tandem predictions. Day 3 illustrates near-saturation sensitivity at Stage 1.

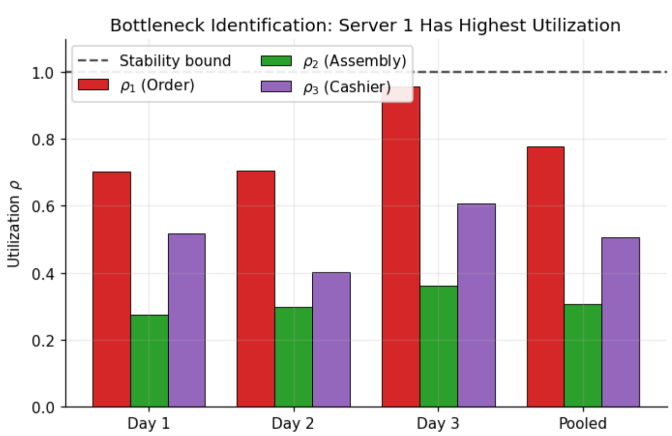

Fig. 3. Estimated utilization by stage. Stage 1 has the highest utilization in every scenario and is close to saturation on Day 3.

VI. Improvement: Parallel Order Taking

A. M/M/2 Stage 1

Since Stage 1 is the bottleneck, the proposed improvement is to add a second order-taker during peak demand. This converts Stage 1 from M/M/1 to M/M/2 while leaving assembly and payment as M/M/1 stages. For an M/M/c queue, define $a = \lambda/\mu$ and $\rho = \lambda/(c\mu)$. The Erlang-C probability that an arrival must wait is

$$C(c,a) = \frac{\frac{a^c}{c!}\frac{1}{1-\rho}}{\sum_{k=0}^{c-1}\frac{a^k}{k!} + \frac{a^c}{c!}\frac{1}{1-\rho}}. \tag{5}$$

Then

$$L_q = C(c,a)\frac{\rho}{1-\rho}, \qquad W_q = \frac{L_q}{\lambda}, \qquad W = W_q + \frac{1}{\mu}. \tag{6}$$

B. Improvement Results

Table IV compares the base and improved designs. In the pooled case, the analytic system time decreases from 7.490 minutes to 3.595 minutes, a 52.0% reduction. Stage 1 alone falls from 5.278 minutes to 1.383 minutes. DES estimates are slightly more conservative but lead to the same decision: system time falls from 7.45 minutes to 4.24 minutes.

TABLE IV
Effect of Adding a Second Order-Taker

Scenario	Base	Improved	DES Improved	Reduction
Day 1	6.502	3.844	4.47 ± 0.04	40.9%
Day 2	5.731	3.119	3.68 ± 0.03	45.6%
Day 3	30.527	4.065	4.86 ± 0.06	86.7%
Pooled	7.490	3.595	4.24 ± 0.04	52.0%

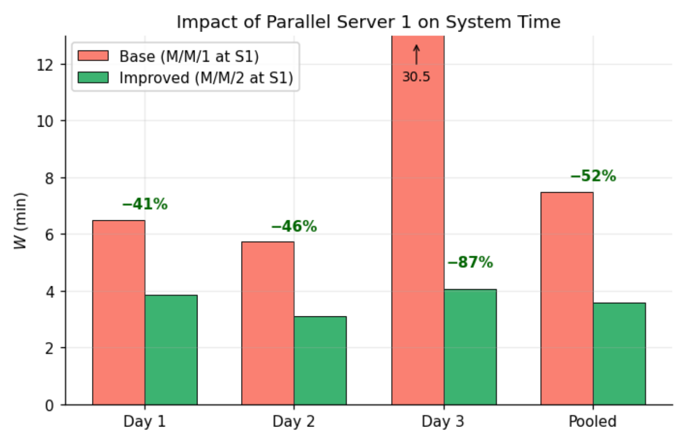

Fig. 4. Analytic mean system time before and after adding a second order-taker. The largest benefit appears on the near-saturated Day 3 scenario.

Sensitivity analysis further supports the intervention. As demand increases, the base M/M/1 Stage 1 approaches instability, causing total system time to grow sharply. The M/M/2 design remains much flatter over the tested demand range, as shown in Fig. 5. This means the second order-taker does more than reduce average wait today; it also makes the system more robust to peak loads.

VII. Limitations and Future Work

The model captures the main flow of the ordering process but simplifies several features of the real restaurant. First, batch arrivals were treated as single entities. This was necessary during data collection, but it hides within-group variation and may understate individual-customer congestion. A future study

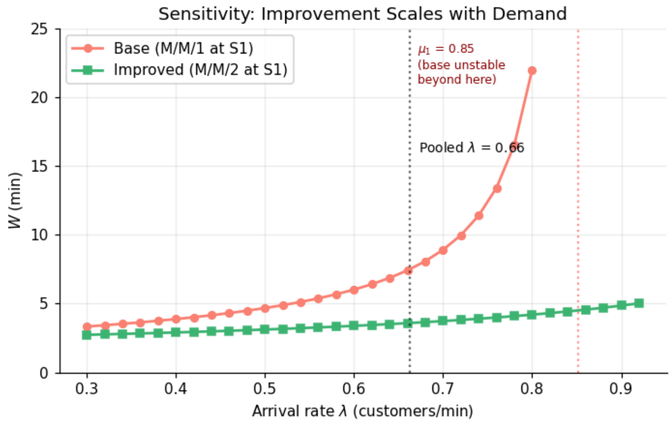

Fig. 5. Sensitivity of system time to arrival rate. The base design grows rapidly as demand approaches Stage 1 capacity, while the M/M/2 Stage 1 design remains stable over the tested range.

should record group size and analyze the system as a batch-arrival queue.

Second, the model assumes a stationary arrival rate. In reality, lunch demand ramps up, peaks, and then tapers off. A time-varying arrival model such as M(t)/M/1 would better represent the lunch window. Third, the model assumes infinite buffers between stages. The actual service line may experience blocking when a downstream worker is unavailable, especially when physical space is limited. Fourth, we model each stage with a single exponential service distribution even though order complexity, worker experience, and customer payment behavior can all change service times.

Finally, the improvement analysis measures time savings but not staffing cost. The recommended next step is a cost-benefit analysis comparing the labor cost of a second order-taker against customer-minutes saved, throughput, and potential sales retained during peak periods.

VIII. Conclusion

This paper modeled Roots Natural Kitchen as a three-stage tandem queueing network and used observed timestamp data to estimate, validate, and improve the system. The pooled model identifies order taking as the bottleneck with utilization 0.778, while assembly and payment have substantially more slack. Analytic Jackson-network results agree closely with discrete-event simulation, giving confidence in the base-model implementation. Adding a second order-taker at Stage 1 is the most direct improvement: it reduces predicted system time by 52.0% in the pooled scenario and prevents the severe waiting-time growth seen near saturation. The analysis therefore supports targeted peak-period staffing at order taking as the clearest operational recommendation.

Acknowledgment

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References

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